EXP NO: 1	Three-Layer Neural Network From Scratch

AIM:

To design and implement a three-layer neural network from scratch using Python, train it with the backpropagation algorithm using suitable activation and loss functions, and apply the model to recognize handwritten digits from the MNIST dataset.

ALGORITHM:

1. Import NumPy and Matplotlib, load the MNIST dataset and use the first 10,000 training images.
2. Normalize the pixel values to the range 0-1, flatten each 28x28 image into 784 features and one-hot encode the labels.
3. Initialize the weights with small random values (x 0.01) and the biases with zeros for the layers 784 - 64 - 10 (input, hidden, output).
4. For every training sample perform forward propagation: z1 = x.W1 + b1, a1 = sigmoid(z1), z2 = a1.W2 + b2, a2 = sigmoid(z2).
5. Compute the cross-entropy loss between the output a2 and the one-hot label y.
6. Perform backpropagation: dz2 = a2 - y, $dW2 = a1^{T}.dz2$, dz1 = $(dz2.W2^{T})$ * a1(1 - a1), $dW1 = x^{T}.dz1$.
7. Update the weights and biases by gradient descent with learning rate 0.1; repeat for 10 epochs and store the average loss of each epoch.
8. Plot the training loss, then predict the digit of a test image using forward propagation and argmax of the output.

CODE:

```
import numpy as np
import matplotlib.pyplot as plt
from tensorflow.keras.datasets import mnist
np.random.seed(42)

# Load MNIST
(X_train, y_train), (X_test, y_test) = mnist.load_data()

# Use a smaller subset so it runs faster on a normal laptop
X_train = X_train[:10000]
y_train = y_train[:10000]

# Normalize and flatten
X_train = X_train.reshape(-1, 784) / 255.0
X_test = X_test.reshape(-1, 784) / 255.0

# One-hot encoding without sklearn version problems
def one_hot(y, classes=10):
    result = np.zeros((len(y), classes))
    result[np.arange(len(y)), y] = 1
    return result
y_train_encoded = one_hot(y_train)

# Network architecture
input_size = 784
hidden_size = 64
output_size = 10
W1 = np.random.randn(input_size, hidden_size) * 0.01
b1 = np.zeros((1, hidden_size))
W2 = np.random.randn(hidden_size, output_size) * 0.01
```

```
b2 = np.zeros((1, output_size))

# Sigmoid
def sigmoid(x):
    x = np.clip(x, -500, 500)
    return 1 / (1 + np.exp(-x))

def sigmoid_derivative(x):
    return x * (1 - x)

# Training
epochs = 10
learning_rate = 0.1
losses = []
for epoch in range(epochs):
    total_loss = 0
    for i in range(len(X_train)):
        x = X_train[i:i+1]
        y = y_train_encoded[i:i+1]
        # Forward propagation
        z1 = np.dot(x, W1) + b1
        a1 = sigmoid(z1)
        z2 = np.dot(a1, W2) + b2
        a2 = sigmoid(z2)
        # Cross entropy loss
        loss = -np.mean(y * np.log(a2 + 1e-8))
        total_loss += loss
        # Backpropagation
        dz2 = a2 - y
        dW2 = np.dot(a1.T, dz2)
        db2 = dz2
        dz1 = np.dot(dz2, W2.T) * sigmoid_derivative(a1)
        dW1 = np.dot(x.T, dz1)
        db1 = dz1
        # Update weights
        W2 -= learning_rate * dW2
        b2 -= learning_rate * db2
        W1 -= learning_rate * dW1
        b1 -= learning_rate * db1
    average_loss = total_loss / len(X_train)
    losses.append(average_loss)
    print("Epoch", epoch + 1, "Loss:", round(average_loss, 4))

# Plot training loss
plt.figure(figsize=(7, 5))
plt.plot(range(1, epochs + 1), losses, marker='o')
plt.xlabel("Epoch")
plt.ylabel("Loss")
plt.title("Training Loss")
plt.grid()
plt.show()

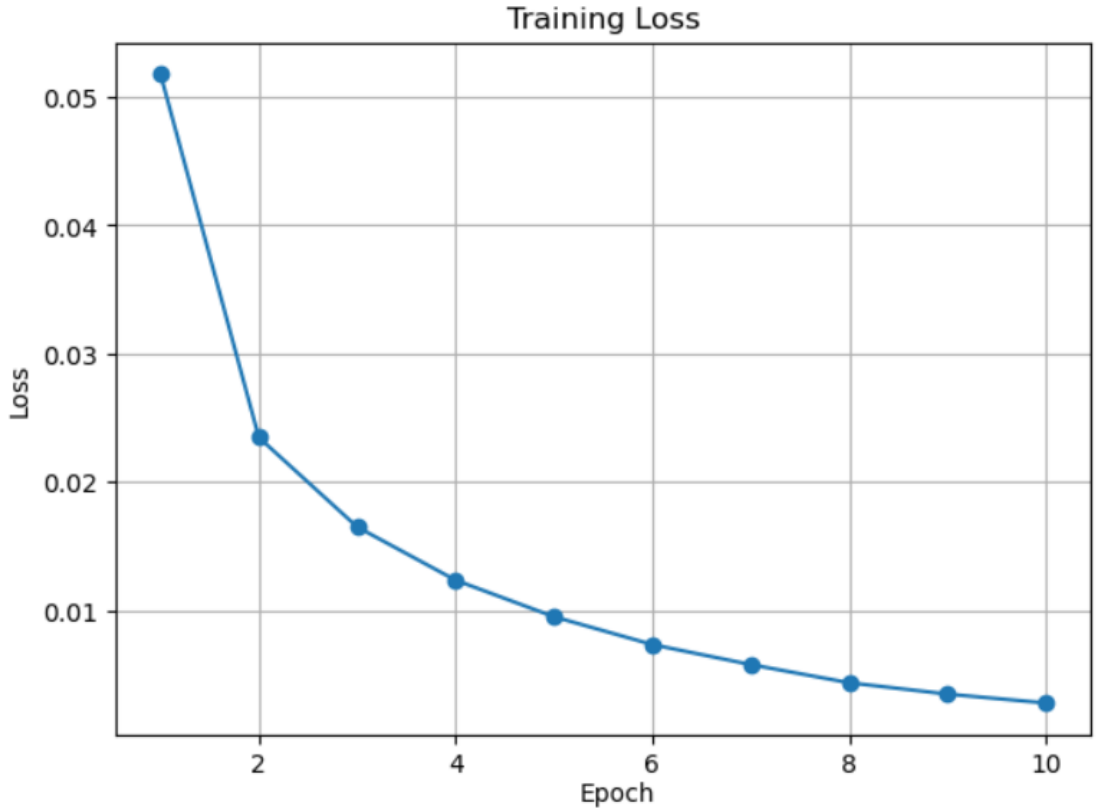
# Prediction
def predict(image):
    image = image.reshape(1, 784) / 255.0
    hidden = sigmoid(np.dot(image, W1) + b1)
    output = sigmoid(np.dot(hidden, W2) + b2)
    return np.argmax(output)

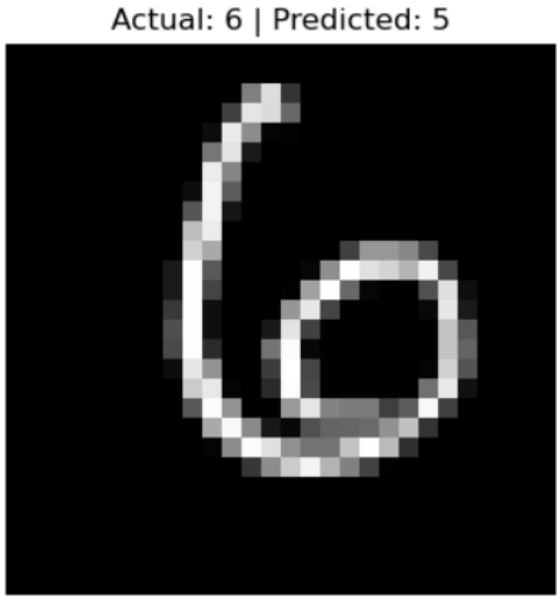
# Test sample
index = 100
prediction = predict(X_test[index])
plt.figure(figsize=(4, 4))
```

```
plt.imshow(X_test[index].reshape(28, 28), cmap="gray")
plt.title("Actual: {} | Predicted: {}".format(y_test[index], prediction))
plt.axis("off")
plt.show()
print("Actual Digit :", y_test[index])
print("Predicted Digit :", prediction)
```

OUTPUT:

```
Epoch 1 Loss: 0.0518
Epoch 2 Loss: 0.0235
Epoch 3 Loss: 0.0165
Epoch 4 Loss: 0.0123
Epoch 5 Loss: 0.0095
Epoch 6 Loss: 0.0073
Epoch 7 Loss: 0.0057
Epoch 8 Loss: 0.0043
Epoch 9 Loss: 0.0034
Epoch 10 Loss: 0.0028
```






```
Actual Digit    : 6
Predicted Digit : 5
```

RESULT:

The three-layer neural network was implemented from scratch and trained with backpropagation on 10,000 MNIST images. The average training loss decreased from 0.0518 in epoch 1 to 0.0028 in epoch 10. For the test image at index 100 the network predicted 5 while the actual digit was 6.